INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

DRILLING MACHINE, RADIAL, FLOOR-MOUNTED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers floor-mounted, radial drilling machines intended for use in general-purpose drilling, tapping, boring, counterboring, spot facing, and reaming operations on large and/or heavy workpieces.

2. CLASSIFICATION. The radial drilling machines shall be of the following sizes and radial arm lengths as shown in table I. The size and radial arm length to be furnished shall be as specified (see 7.2(b)).

3. SALIENT CHARACTERISTICS

3.1 General requirements. The radial drilling machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The radial drilling machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the machine. The top of the base shall be machined as a work surface.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3413

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TABLE I. Sizes and radial arm lengths of drilling machines.

Size[1]	Maximum drill diameter, inches[2]	Radial arm length, inches[3]													
		25	30	35	40	45	50	60	65	70	80	85	95	100	120
01	1 1/4				X										
02	2	X	X	X	X	X	X	X							
03	2 1/4			X		X		X							
04	2 3/8		X		X	X	X	X	X						
05	2 1/2					X	X	X			X				
06	2 5/8						X	X	X						
07	3							X	X		X	X			
08	3 1/2						X		X		X			X	X
09	4						X	X		X	X		X	X	X
10	4 3/4														X

[1]The machine maximum drill diameter and radial arm length shall not be less than the stated requirements of the size ordered nor greater than the next larger size shown.
[2]The maximum drill diameter is the machine rating when drilling cast iron. The maximum drill diameter is less when drilling steel.
[3]The radial arm length of the drilling machine is defined as the horizontal distance between the column face and the centerline of the drill head spindle at its maximum travel distance from the column.

3.2.2 Table. Unless otherwise specified (see 7.2(c)), the radial drilling machine shall be supplied with a plain box table. The table shall have at least two machined work surfaces that are adjacent at 90 degrees to each other. A means to attach the table to the base shall be provided. Unless otherwise specified (see 7.2(d)), the table dimensions and number of work surfaces shall be that considered by the manufacturer as standard.

3.2.3 T-slots. The number and size of the base and table T-slots shall be as specified (see 7.2(e)). The dimensions of the T-slots shall conform to the requirements of American National Standards Institute (ANSI) B5.1M, "T-slots, Their Bolts, Nuts, and Tongues".

3.2.4 Column. The column shall be designed to resist forces developed during drilling operations and minimize vibration. The column shall pivot on the base or the drill head radial arm shall pivot about the column to provide angular motion of the drill head assembly. Unless otherwise specified (see 7.2(f)), the column shall have automatic lubrication and the drill head radial arm shall be equipped with a wiping system to keep the column clean.

3.2.5 Radial arm. The radial arm shall provide rigid support to the drill head assembly, power transmission, and drive motor. The radial arm shall resist forces and minimize vibration. The radial arm shall be vertically adjustable on the column and have a means for clamping the arm to the column after adjustment. Unless otherwise specified (see 7.2(g)), the radial arm shall rotate in a horizontal plane a minimum of 135 degrees left and right of center. Unless otherwise specified (see 7.2(h)), safety devices shall be provided to prevent vertical radial arm overtravel and, if equipped with a screw and nut drive system, to prevent unintended downward radial arm movement because of screw and nut failure.

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3.2.6 Drill head assembly. The drill head assembly shall move horizontally along the radial arm. Means shall be provided to clamp the drill head assembly to the arm after adjustment.

3.2.6.1 Spindle and quill. Unless otherwise specified (see 7.2(i)), the spindle shall be provided with the taper considered by the manufacturer as standard. Unless otherwise specified (see 7.2(j)), the spindle shall be provided with manual feed, adjustable power feed, and an adjustable depth stop. Power feed shall be disengaged automatically when the depth stop is reached. The operator shall be able to manually position the spindle and quill and then engage the power feed. The spindle and quill shall have a counterweight, counterbalance, or both to automatically return the spindle and quill to its highest position when the manual feed lever is released or the power feed is disengaged.

3.2.6.2 Motor and power transmission. The machine shall have a variable speed motor or have a transmission that allows spindle speed adjustment. Unless otherwise specified (see 7.2(k)), a clutch shall be provided to prevent motor damage during an overload.

3.2.7 Clamping system. The clamping system for the radial arm, drill head assembly, and column shall be powered either by an electric, hydraulic, or combination system. The clamping system shall be provided with interlocks to prevent powered movement while the clamp is engaged. During a loss of machine power, the clamping system shall not unclamp.

3.2.8 Controls. The controls shall be located for convenient access by the operator. Unless otherwise specified (see 7.2(l)), the following switches and controls, as a minimum, shall be provided:

a. On/off power switch
b. Emergency stop button
c. Column movement
d. Vertical radial arm movement
e. Horizontal drill head assembly movement
f. Power feed
g. Spindle speed
h. Coolant flow
i. Spindle rotation direction selector

3.2.9 Lubrication system. Unless otherwise specified (see 7.2(m)), the drilling machine shall be equipped with an automatic lubricating system to direct oil flow from the reservoir to all machine components requiring continuous lubrication. A means shall be provided for determining oil level in the reservoir. The automatic lubricating system shall include a cleanable or replaceable filter. Upon loss of pressure in the automatic lubrication system, the drilling machine shall automatically stop. Machine parts requiring periodic manual lubrication shall be readily accessible through doors or removable panels.

3.2.10 Coolant system. Unless otherwise specified (see 7.2(n)), the drilling machine shall be flood cooled. As a minimum, the coolant system shall consist of a reservoir, pump, strainer, and all necessary piping. The coolant system shall be equipped with a method to control the rate of coolant flow from the reservoir and to direct coolant flow to the drilling area. When specified (see 7.2(o)), an optional filtration system shall be provided.

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3.2.11 Hydraulic system. If a hydraulic system is provided, it shall conform to the requirements of National Fluid Power Association (NFPA) T2.24.1, "Hydraulic Fluid Power - Systems Standard for Stationary Industrial Machinery" (DoD adopted). The hydraulic system shall be complete and include all pumps, valves, piping, cylinders, fluid level indicator, and controls. Means shall be provided to prevent damage to the components from overpressure. A filter system shall be provided to ensure delivery of clean fluid.

3.2.12 Electrical system. Unless otherwise specified (see 7.2(p)), the electrical system shall conform to the National Fire Prevention Association (NFPA) 79, "Electrical Standard for Industrial Machinery" (DoD adopted). The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(q)). Unless otherwise specified (see 7.2(r)), the control system power shall be provided by a separate 120-volt or less, single-phase, 60-hertz circuit from the machine main power supply.

3.2.13 Electric motors. The drill head motor and auxiliary (if applicable) drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) MG 1, "Motors and Generators" (DoD adpoted). Unless otherwise specified (see 7.2(s)), the motors shall have open dripproof enclosures. Each motor control shall conform to NEMA ICS 1, "Industrial Control and Systems General Requirements" (DoD adpoted), and NEMA ICS 3, "Industrial Systems" (DoD adpoted).

3.2.14 Optional accessories. Optional accessories shall be provided as specified (see 7.2(t)).

3.3 Performance. Unless otherwise specified (see 7.2(u)), the radial drilling machine shall meet the alignment and performance requirements shown in tables I and II, respectively, when tested in accordance with sections 5.3 through 5.3.5.

TABLE II. Alignment requirements.

Parameter	Tolerance
Work surface, flat in longitudinal direction	0.0012 inch per foot, TIR[1]
Work surface, flat in traverse direction	0.0012 inch per foot, TIR
Column square with base in plane through base centerline: for machines up to a 48-inch radial arm length for machines over a 48-inch radial arm length	 0.0024 inch per foot, TIR 0.0036 inch per foot, TIR
Column square with base in plane perpendicular to base centerline	0.0012 inch per foot, TIR
Radial arm parallel with base: for machines up to a 48-inch radial arm length for machines over a 48-inch radial arm length	 0.0024 inch per foot, TIR 0.0036 inch per foot, TIR
Spindle runout	0.002 inch per foot, TIR
Spindle square with base in plane through base centerline	0.0024 inch per foot, TIR
Spindle square with work surface of base in plane 90 degrees to base centerline	0.0012 inch per foot, TIR
Travel of quill square with work surface of base in plane through base centerline	0.004 inch per foot, TIR
Travel of quill square with base in plane 90 degrees to base centerline	0.002 inch per foot, TIR

[1]TIR is the total indicated reading.

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TABLE III. Performance requirements.

Parameter	Tolerance
Vertical alignment	0.01 inch/foot, TIR[1]
Hole straightness	0.005 inch, TIR hole top to bottom
Hole maximum diameter	Test drill diameter + 0.007 inch
Hole diameter out of roundness	0.005 inch, TIR

[1]TIR is the total indicated reading.

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(v)).

3.5 Safety and health requirements. The machine shall be designed and manufactured in accordance with ANSI B11.8, "Drilling, Milling, and Boring Machines" (DoD adpoted). The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with Occupational Safety and Health Administration's (OSHA) Title 29 of the Code of Federal Regulations (CFR), Part 1910, "Occupational Safety and Health Standards". If a conflict arises between the OSHA and ANSI standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(w)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(x)), measuring and indicating devices shall be graduated in both the US and SI system of measurements.

3.7 Nameplate. A nameplate shall be securely attached to the machine. Unless otherwise specified (see 7.2(y)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer's name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, and frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(z)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet all applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Inspection. The radial drilling machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(aa)), the radial drilling machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(bb)), the manufacturer shall be responsible for supplying all tooling and material(s) and shall conduct all tests required by the Government. The manufacturer shall provide test instruments and equipment with calibration traceable to National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(cc)). Unless otherwise specified (see 7.2(dd)), the machine shall be tested in accordance with the following environmental conditions: the ambient temperature shall be between 41 and 104 °F, and the humidity shall be between 20 and 95 percent, non-condensing.

5.3.2 Operational test. The radial drilling machine shall be operated in accordance with the manufacturer's standard operational test procedures. The radial drilling machine shall be operated at no load for not less than 1 hour. Proper operation of all controls and features shall be verified during the trial period.

5.3.3 Alignment tests. The radial drilling machine alignment tests shall be conducted before and after the performance tests and shall meet the requirements shown in table II. Changes in alignment shall be cause for rejection.

5.3.3.1 Work surface inspection. The work surfaces of the base and table shall be measured for flatness. The table shall be mounted and secured to the base during inspection.

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5.3.3.2 Column and base squareness. The drill head assembly shall be positioned at the maximum distance from the column, and the radial arm raised to the maximum height. The column squareness with the base and radial arm parallelism to the base shall be measured.

5.3.3.3 Spindle runout. With the drill head assembly and the drill head height set at the manufacturer's discretion, the spindle runout shall be measured. The spindle and quill shall be fully retracted. A mandrel extending 12 inches from the spindle nose shall be used. The measurement of spindle runout shall be taken at the end of the mandrel.

5.3.3.4 Spindle and base squareness. The radial arm height shall be set at the midpoint of arm travel and the drill head assembly positioned at the maximum distance from the column. The spindle and quill travel squareness with the base shall be measured. Spindle centerline tilt toward the column shall be considered positive, and spindle centerline tilt away from the column shall be considered negative.

5.3.4 Performance tests. The performance tests shall be conducted in accordance with sections 5.3.1 through 5.3.2. There shall be no evidence of overloading, overheating, excessive vibration, or deformation of any part of the machine during or after the test.

5.3.4.1 Test workpiece. A test workpiece 1.0 inch thick and made of soft cast iron with a Brinell hardness of 120-150 shall be used. Other dimensions are at the manufacturer's discretion. The minimum size of the test workpiece shall allow for all of the required tests. Ten holes shall be drilled through the test workpiece. All drilling shall be finished in 30 minutes. After drilling five holes, the drill bit may be replaced. The time used to replace the bit shall not count towards the 30-minute time limit. The drill bit nominal diameter shall be the maximum rated drilling diameter for the machine. Drill bit material, drill spindle speed, and feed rate shall be at the option of the manufacturer. The holes shall be measured and shall meet the requirements shown in table III for vertical alignment, hole straightness, maximum hole diameter, and out of roundness. The diameter of all holes shall be measured at the top, midpoint, and bottom. Out of roundness, the difference between the largest diameter of the hole and smallest diameter of the hole, shall be determined by moving the measuring device 360 degrees around the drilled hole.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(ee)).

5.4 Acceptance. Unless otherwise specified (see 7.2(ff)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables II and III, respectively. Failure of the machine to meet the requirements shown in tables II and III shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the radial drilling machine shall be preserved, packed, and marked as specified in the acquisition order (see 7.2(gg)).

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7. NOTES

7.1 Sources of documents.

7.1.1 Government documents. Copies of CFR and FAR may be obtained from the U.S. Government Printing Office, Superintendent of Documents, Mail Stop: SSOP, Washington, DC 20402-9328. Electronic copies of CFR documents may be obtained from http://www.access.gpo.gov.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.4 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

7.1.5 NFPA standards. Copies of NFPA standards may be obtained from the National Fluid Power Association, 3333 North Mayfair Road, Suite 311, Milwaukee, WI 53222-3219.

7.2 Ordering data. Acquisition documents must specify the following:

a. CID document number, revision, and CID PIN
b. Size and radial arm length of drilling machine required (see 2)
c. Table, if different (see 3.2.2)
d. Table dimensions and number of work surfaces, if different (see 3.2.2)
e. Number and size of T-slots, base and table, as specified (see 3.2.3)
f. Automatic column lubrication system, if different (see 3.2.4)
g. Radial arm angular rotation, if different (see 3.2.5)
h. Radial arm safety devices, if different (see 3.2.5)
i. Spindle taper, if different (see 3.2.6.1)
j. Spindle feed and depth stop, if different (see 3.2.6.1)
k. Clutch, if different (see 3.2.6.2)
l. Controls, if different (see 3.2.8)
m. Lubrication system, if different (see 3.2.9)
n. Coolant system, if different (see 3.2.10)
o. Coolant filtration system, if required (see 3.2.10)
p. Electrical system industrial standard, if different, (see 3.2.12)
q. Electrical system voltage, as specified (see 3.2.12)
r. Control system voltage, if different (see 3.2.12)
s. Motor enclosure, if different (see 3.2.13)
t. Optional accessories, as specified (see 3.2.14)
u. Performance, if different (see 3.3)
v. Machine dimensions and weight, as specified (see 3.4)
w. System of units required (see 3.6)
x. Dual system of units, if required (see 3.6)

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y. Nameplate, if different (see 3.7)
z. Lubrication plate or chart, if required (see 3.8)
aa. Testing, if different (see 5.3)
bb. Test responsibility, if different (see 5.3.1)
cc. Certification of calibration, as specified (5.3.1)
dd. Test environmental conditions, if different (see 5.3.1)
ee. Optional tests, as specified (see 5.3.5)
ff. Acceptance, if different (see 5.4)
gg. Packaging requirements, if different (see 6.1)

7.3 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

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Radial arm length:

NOTE:
See table 1 for applicable radial arm length.

A - 25 inches
B - 30 inches
C - 35 inches
D - 40 inches
E - 45 inches
F - 50 inches
G - 60 inches
H - 65 inches
I - 70 inches
J - 80 inches
K - 85 inches
L - 95 inches
M - 100 inches
N - 120 inches

Size: See table I

CID number

MILITARY INTERESTS:

Custodians:
Army - AL
Navy - SH
Air Force - 99

Reviewers:
Army - SM
Navy - MC
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

(Project 3413-0166)

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